ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** vibrating bodies	Vibration is the source.
2	**(c)** amplitude	Larger amplitude, louder.
3	**(b)** 17.2 m	v t/2 = 344 x 0.1/2.
4	**(c)** a vacuum	A material medium is required.
5	**(b)** frequency	Higher frequency, higher pitch.
6	**(c)** echolocation using ultrasonic waves	Reflected ultrasound.
7	**(d)** longitudinal waves	Particles vibrate along the direction of travel.
8	**(b)** distinguish two sounds of the same pitch and loudness	Waveform differs.
9	**(c)** sound navigation and ranging	Uses ultrasonic waves.
10	**(b)** compression	High pressure region.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	By vibrating objects (1); A material medium (solid, liquid or gas) (1).	1+1
12	Pitch depends on frequency (1); Loudness depends on amplitude (1).	1+1
13	20 Hz to 20 kHz (1); Below 20 Hz and above 20 kHz respectively (1).	1+1
14	Particles of the medium vibrate parallel to the direction of the wave (1); Compressions and rarefactions travel (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	v = f lambda = 340 m/s (1); T = 1/200 (1); = 0.005 s (1).	1+1+1
16	Pulse sent and its echo is received (1); Depth = v t/2 (1); = 1500 x 4/2 = 3000 m (1).	1+1+1
17	Speed increases with temperature (1); Particles move faster and transfer disturbance quickly (1); About 0.6 m/s per degree Celsius (1).	1+1+1
	OR: lambda1 = 344/20 = 17.2 m (1); lambda2 = 344/20000 (1); = 0.0172 m = 1.72 cm (1).	1+1+1
18	Pitch: frequency (1); Loudness: amplitude (1); Quality/timbre: waveform (1).	1+1+1
	OR: Wavelength = 0.68 m (1); 500 waves in 1 s (1); 10/0.68 = 14.7 waves (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Vibrating source (1); Compression and rarefaction (1); Amplitude and wavelength (1); Frequency and time period f = 1/T (1); Wave equation v = f lambda (1).	1+1+1+1+1
	OR: Repeated reflections (1); Persists and blurs speech (1); Absorbing materials: curtains, carpets (1); Curved sound boards reflect sound to the audience (1); Roof design (1).	1+1+1+1+1
20	v = distance/time = lambda/T (1); = lambda f (1); v = 256 x 1.34 (1); = 343 m/s (1); Units (1).	1+1+1+1+1
	OR: Depth = 1500 x 2.4/2 = 1800 m (1); Echo time for 1650 m = 3300/1500 (1); = 2.2 s (1); Formula d = vt/2 (1); Reason for halving (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.

- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 20 Hz to 20000 Hz	Normal hearing.
2	**(c)** find the depth of the sea	Echo ranging.
3	**(a)** rarefaction	Low pressure region.
4	**(d)** infrasonic	Below the audible range.
5	**(c)** imaging internal organs	Safe imaging.
6	**(b)** wavelength	Distance for one cycle.
7	**(d)** ultrasonic	Above the audible range.
8	**(b)** reverberation	Sound persists.
9	**(d)** frequency	Hz.
10	**(a)** solids	Particles are closest.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sound heard after reflection from a surface (1); The time gap should be at least 0.1 s (1).	1+1
12	Medical imaging (1); Cleaning and SONAR (1).	1+1
13	Frequency: oscillations per second (1); Wavelength: distance for one cycle; amplitude: maximum displacement (1).	1+1
14	Finding objects using reflected ultrasonic waves (1); Bats and dolphins (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Strike a tuning fork on a rubber pad (1); Touch a light pith ball with it (1); The ball is thrown away, showing vibration (1).	1+1+1
16	Particles are closer in solids (1); Air about 340, water about 1500 (1); Steel about 5000 m/s (1).	1+1+1
17	Transverse: particles vibrate perpendicular, e.g. wave on a rope (1); Longitudinal: parallel, e.g. sound (1); Compressions vs crests (1).	1+1+1
	OR: Vibrating source pushes air layers: compression (1); Pulls back: rarefaction (1); The disturbance travels, not the air itself (1).	1+1+1
18	Light travels much faster than sound (1); Distance = 340 x 5 (1); = 1700 m (1).	1+1+1
	OR: Sine curve (1); Crest: highest point; trough: lowest (1); Amplitude: height from mean; wavelength: distance between two crests (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pitch: frequency (1); Loudness: amplitude (1); Quality: waveform (1); Higher frequency means closer waves (1); Larger amplitude means taller waves (1).	1+1+1+1+1
	OR: Solids fastest (steel about 5000 m/s) (1); Liquids intermediate (water about 1500 m/s) (1); Gases slowest (air about 340 m/s) (1); Increases with temperature (1); Particle spacing and elasticity (1).	1+1+1+1+1
20	Echo: reflected sound (1); t = 170/340 = 0.5 s (1); Heard distinctly as t > 0.1 s (1); Minimum distance = 17 m (1); Persistence of hearing 0.1 s (1).	1+1+1+1+1
	OR: Electric bell in a bell jar (1); Pump out the air (1); Sound becomes fainter and then inaudible (1); The hammer is still seen striking (1); Sound cannot travel in a vacuum (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.

- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.
- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** absorb sound and reduce reverberation	Soft materials absorb.
2	**(c)** hertz	1 Hz = 1 oscillation per second.
3	**(d)** 344 m/s	About 340 m/s.
4	**(a)** multiple reflections of sound	Guide sound.
5	**(d)** f = 1/T	Reciprocals.
6	**(b)** temperature	Warmer air.
7	**(b)** 340 m/s	f lambda.
8	**(c)** v = f lambda	Wave equation.
9	**(b)** 0.1 s	Persistence of hearing.
10	**(c)** 0.02 s	1/f.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Persistence of sound due to repeated reflections (1); Using sound-absorbing curtains and carpets (1).	1+1
12	f = 1/T (1); T = 0.2 s (1).	1+1
13	No atmosphere (no medium) on the Moon (1); Sound needs a medium (1).	1+1
14	lambda = v/f (1); = 2 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total distance = 344 x 3 = 1032 m (1); Distance of surface = 516 m (1); Sound travels to and fro (1).	1+1+1
16	Sound of the heartbeat enters the chest piece (1); Reflected repeatedly along the tube (1); Reaches the doctor's ears with less loss (1).	1+1+1
17	Energy spreads over a larger area (1); Amplitude decreases with distance (1); Loudness decreases (1).	1+1+1
	OR: Bats emit ultrasonic waves (1); Waves reflect from the insect (1); The bat detects the echo and finds position (1).	1+1+1
18	Imaging of the foetus (1); Cleaning of delicate parts (1); Detecting cracks in metals (1).	1+1+1
	OR: Sound needs a medium; light does not (1); Sound is longitudinal; light is transverse (1); Sound is much slower (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pinna collects sound (1); Eardrum vibrates (1); Ossicles amplify (1); Cochlea converts to electrical signals (1); Auditory nerve to the brain (1).	1+1+1+1+1
	OR: Audible 20 Hz to 20 kHz (1); Infrasonic: elephants, earthquakes (1); Ultrasonic: bats, dolphins (1); Dog whistle (1); Uses of ultrasound (1).	1+1+1+1+1
20	Ultrasound is reflected (1); Depth = v t/2 (1); Imaging of organs (1); Cleaning and flaw detection (1); Why ultrasonic waves are used: short wavelength, directional (1).	1+1+1+1+1
	OR: Unwanted loud sound (1); Sources: traffic, loudspeakers, machines (1); Effects: hearing loss, stress (1); Control: silencers, trees, rules (1); Use of ear protection (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
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